

Table 1: BPE-token Jaccard similarity between APC (direct sampling + Adaptive Plausibility Constraint) and Greedy Search on LLaVA-Bench (In-the-Wild), evaluated on Qwen2.5-VL-7B. Jaccard scores are computed over sets of unique BPE token IDs using the Qwen2 tiktoken tokenizer (vocabulary size 151,643) and averaged across all 60 questions. Text is normalised before tokenisation to remove markdown artefacts (****bold****, headers, bullets) introduced by the model despite a no-formatting system prompt. Higher values indicate greater overlap with greedy search.

APC β	Mean Jaccard (BPE, Qwen2 tiktoken)
0.000	0.3713
0.025	0.3710
0.050	0.4327
0.075	0.4388
0.100	0.4474
0.200	0.4763
0.300	0.4850
0.400	0.4943
0.500	0.5219
0.600	0.5506
0.700	0.5757
0.800	0.6192
0.900	0.7308
1.000	0.7785

Table 2: BPE-token Jaccard similarity of APC outputs against Greedy Search and Direct Sampling at four representative β values, evaluated on Qwen2.5-VL-7B across 60 LLaVA-Bench (In-the-Wild) questions. Tokenisation uses the Qwen2 tiktoken tokenizer (vocabulary size 151,643). In every case, APC outputs are more similar to Greedy Search than to Direct Sampling, showing that even at low β , APC pulls the output distribution towards greedy rather than preserving the diversity of direct sampling.

APC β	Jaccard vs. Greedy	Jaccard vs. Direct Sampling
0.075	0.4388	0.3427
0.100	0.4474	0.3445
0.200	0.4763	0.3702
0.300	0.4850	0.3661

Claim: When direct sampling is combined with the Adaptive Plausibility Constraint (APC), the outputs become more similar to greedy search than to direct sampling itself, even at small values of β . This shows that APC does not preserve the diversity of direct sampling — instead, it systematically shifts the output distribution towards the deterministic greedy solution.

Table 3: BPE-token Jaccard similarity between VCD (Visual Contrastive Decoding) outputs and three reference conditions — Greedy Search, APC at its default $\beta=0.1$, and Direct Sampling — evaluated on Qwen2.5-VL-7B across 42 LLaVA-Bench (In-the-Wild) questions (42 of 60 completed). Tokenisation uses the Qwen2 tiktoken tokenizer (vocabulary size 151,643).

Comparison	Mean Jaccard (BPE, Qwen2 tiktoken)
VCD vs. Greedy Search	0.4305
VCD vs. APC ($\beta = 0.1$)	0.3974
VCD vs. Direct Sampling	0.3587

Explanation: VCD outputs are most similar to Greedy Search (Jaccard = 0.4305) and least similar to Direct Sampling (Jaccard = 0.3587), even though VCD is built on top of direct sampling. This happens because VCD internally applies APC at $\beta=0.1$ to filter the output distribution at every decoding step. As shown in Table ??, APC shifts direct sampling towards greedy search even at small β values. Since VCD inherits this APC step, its outputs end up closer to greedy search than to the direct sampling distribution it was intended to improve upon.